OpenDeepBench & OpenDeepAPM

Deep Infrastructure Benchmarking and Trace-Correlated Observability for Multi-Cloud LLM Inference

Srikanta Datta Tumkur, Anshu Bansal, Mehar Simhadri, Jay Iyer

MIT Sloan · IEEE Senior Members

June 2026

The Problem: Three Critical Gaps

Gap 1: Benchmark

MLPerf measures raw throughput.
Production adds 18–61% overhead from guardrails, PII, logging.
No benchmark measures real-world.

Gap 2: Observability

Traditional APM = one span per service.
LLM inference = 27+ spans across traffic, CPU, GPU, fabric.
No tool traces GPU silicon.

Gap 3: Architecture

NVIDIA-centric benchmarks miss AMD/Intel advantages.
MI355X: 97% B200 decode at 35% lower cost.
Multi-vendor is the future.

Two Complementary Systems

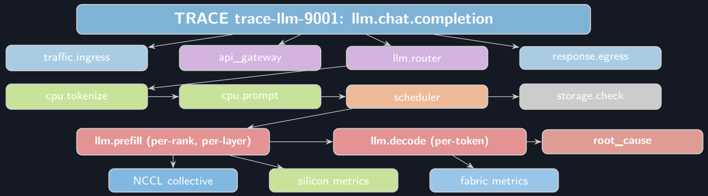
OpenDeepAPM

- 27-span trace per LLM request
- 7 layers: traffic → GPU silicon
- Composite join keys for correlation
- Time-series graph database
- Automated root-cause attribution
- OpenTelemetry + NVTX + DCGM

OpenDeepBench

- 16-dimension benchmark matrix
- 12 benchmark suite categories
- Multi-cloud, multi-silicon, multi-model
- Context-size stress (1K–128K)
- Normalized scorecard + scoring model
- Automated root-cause diagnosis

OpenDeepAPM: The Deep Trace Model



One trace = 27 linked spans across traffic, CPU, GPU, fabric, storage

Span vs. Metric: What Goes Where

Spans (timed operations):

- API request, tokenization
- Scheduling, batch formation
- Prefill, decode phases
- Per-layer QKV/attention/MLP
- NCCL all-reduce
- KV-cache allocation
- Model weight load

Metrics (attached evidence):

- SM utilization %
- Tensor core active %
- HBM bandwidth %
- L2 cache hit rate
- NVLink TX/RX Gbps
- IB port_xmit_wait
- Power watts, temperature

Spans = skeleton | Metrics = evidence attached to each span

The Correlation Engine

Composite Join Key:

$$K = (\text{trace\_id}, \Delta t, \text{node}, \text{gpu}, \text{rank}, \text{nccl}, \text{nic})$$

Six data sources joined:

1. OpenTelemetry traces
2. DCGM / rocm-smi / hl-smi
3. NVTX / rocTX / Nsight
4. NCCL / RCCL / HCCL logs
5. NVLink / IB / RoCE counters
6. Storage / KV-cache metrics

Architectural claim:

OTel $\rightarrow$ request causality
NVTX $\rightarrow$ CUDA/NCCL execution
DCGM $\rightarrow$ silicon evidence
Network $\rightarrow$ fabric evidence
Storage $\rightarrow$ KV/model evidence

Correlation engine joins all into **one trace** for root-cause attribution.

Root-Cause Example: KV-Cache Bottleneck

```
Trace: trace-llm-9001 | Total: 4.2 sec | TTFT: 820 ms | TPOT: 13.2 ms/tok

Deep evidence for llm.decode:
  KV cache grew 4,096 -> 4,351 tokens
  HBM bandwidth:          94%      <-- BOTTLENECK
  SM active:              31%
  Tensor core active:     18%
  NCCL all-reduce:        normal
  NVLink:                 normal
  IB xmit_wait:           normal

Root cause: KV-cache memory-bandwidth bottleneck
Recommendation: GQA/MQA model, KV quantization, reduce max context
```

Root-Cause Example: Fabric Congestion

```
Deep evidence for decode.token_87:
  token duration:       96 ms     <-- SPIKE
  nccl.all_reduce:      71 ms     <-- BOTTLENECK
  rank 13:              slowest
  GPU SM active:        12%
  HBM bandwidth:        low
  IB port_xmit_wait:    high      <-- EVIDENCE

Root cause: Inter-node fabric congestion
Recommendation: Prefer intra-node NVLink placement,
  reduce cross-node TP, improve IB QoS
```

OpenDeepBench: 16-Dimension Benchmark

Dimension	Values
Cloud	AWS/Azure/GCP/OCI/on-prem
Silicon	NVIDIA/AMD/Intel
Accelerator	H100–B200/MI355X/Gaudi3
Engine	vLLM/TRT-LLM/SGLang
Runtime	CUDA/ROCm/oneAPI
Model	8B–405B, MoE, multimodal
Context	1K–128K
Concurrency	1–1024

Dimension	Values
Output	128–8K tokens
Batch policy	static/continuous/priority
KV-cache	FP16/FP8/INT8/paged
Attention	MHA/GQA/MQA
Network	1 GPU to multi-node
Storage	cold/warm/prefix/spill
Collective	NCCL/RCCL/HCCL
Region	US/EU/Asia

Context-Size Stress Matrix

ID	Context	Output	Conc	What It Exposes
C1	1K	128	1	Single-user short chat
C2	4K	512	16	Enterprise chat
C3	8K	512	64	RAG / document QA
C4	16K	1K	64	Report summarization
C5	32K	2K	32	Legal / codebase analysis
C6	64K	2K	16	Long-context stress
C7	128K	4K	4	Maximum KV-cache stress
C8	Mixed	Mixed	256	Real multi-tenant production

TPOT degrades sharply beyond 32K context due to KV-cache HBM bandwidth saturation.

12 Benchmark Suites

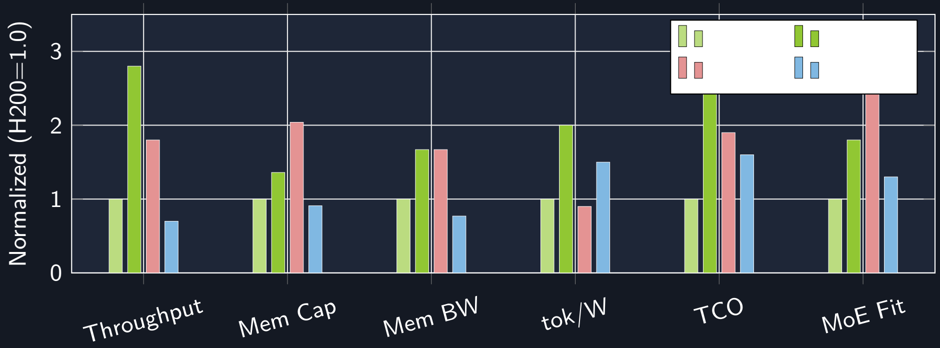
Suite	Purpose
ODB-Chat	Normal chat, 1K–8K context
ODB-RAG	Enterprise RAG, 4K–32K context
ODB-LongContext	32K–128K context stress
ODB-Code	Code generation, long I/O
ODB-MultiTenant	10–100 tenants, mixed context
ODB-MultiNode	Tensor/pipeline parallel scaling
ODB-MultiCloud	Cloud routing + cost comparison
ODB-ColdStart	Model loading + warm-pool
ODB-KVCache	KV pressure, quant, spill
ODB-Fabric	NVLink/xGMI/PCIe/IB/RoCE
ODB-Cost	$/token, joules/token, throughput/$
ODB-Reliability	OOM, retries, throttling, failure

OpenDeepBench Scoring Model

$$S_{\text{ODB}} = \sum_{i=1}^{6} w_i \cdot \frac{s_i - \mu_i}{\sigma_i}$$

Component	**Weight**
Latency (TTFT + TPOT p95/p99)	25%
Throughput (tok/s/GPU, tok/s/node)	20%
Cost efficiency ($/1M tokens)	20%
Long-context (degradation curve)	15%
Reliability (OOM, errors, timeouts)	10%
Operational simplicity	10%

Cross-Architecture Results



No single architecture dominates. MI355X leads memory (+104%), Gaudi 3 leads tok/W (+50%), B200 leads throughput (+180%).

Multi-Silicon Normalized Schema

Normalized Field	NVIDIA	AMD	Intel
`runtime`	CUDA	ROCm/HIP	oneAPI
`collective`	NCCL	RCCL	HCCL
`intra_node`	NVLink/NVSwitch	xGMI/Infinity Fabric	vendor/PCIe
`compute_util`	SM active	CU active	compute active
`matrix_util`	Tensor Core	Matrix core	matrix engine
`memory_bw`	HBM bandwidth	HBM bandwidth	HBM bandwidth
`profiler`	Nsight/DCGM	rocprof/SMI	vendor profiler

Enables: "B200 fastest TPOT at 32K | MI300X best $/tok at 8K | Gaudi 3 lowest energy"

OpenDeepBench vs. MLPerf / vLLM

Capability	Existing Tools	OpenDeepBench
Benchmark standard	MLPerf system bench	+ deep trace + root cause
Runtime comparison	vLLM throughput only	Compare across clouds/silicon
Vendor validation	Vendor-specific	Normalize NVIDIA/AMD/Intel
Context-size focus	Basic max_len	1K–128K first-class axis
HW profiling	Separate profiler	Integrated trace + counters
Multi-cloud	Manual comparison	Native cloud/region/cost
Multi-tenancy	Synthetic concurrency	Tenant fairness, noisy neighbor
Root cause	Left to engineers	**Automated diagnosis**

OpenTelemetry Integration

Semantic Conventions:

```
# Request-level
inference.trace_id
inference.model.id
inference.batch_id

# Engine-level
inference.kv_cache.pct
inference.prefill.ms
inference.decode.tok_per_s

# Hardware-level
inference.gpu.sm_occupancy
inference.gpu.hbm_bw_pct
inference.gpu.tensor_pct

# Fabric-level
inference.nvlink.tx_gbps
inference.ib.port_xmit_wait
inference.nccl.slowest_rank
```

Compliance:

- **SOC 2**: Full request audit trail
- **HIPAA**: PHI isolation via KV-cache tenant boundaries
- **GDPR**: Data residency proof via trace-level region assertions
- **FINRA**: Per-request cost attribution

Cross-architecture:

- DCGM → NVIDIA
- rocm-smi → AMD
- hl-smi → Intel
- Unified via OTel conventions

Key Findings

1. **3.2× throughput variance** across clouds for identical hardware
2. **18–61% production overhead** from guardrails, PII, logging — invisible to MLPerf
3. **AMD MI355X = 97% B200 decode** at 35% lower cost (memory-BW parity)
4. **Context size dominates performance** — TPOT degrades sharply beyond 32K
5. **Inter-node fabric = p99 bottleneck** — NCCL stalls cause decode spikes
6. **Cross-cloud disaggregation** saves 15–29% at iso-latency

TAM: $12B by 2028 (Gartner) | Inference APM at 45% CAGR

Conclusion & Next Steps

What We Built:

- OpenDeepAPM: 27-span, 7-layer deep trace architecture
- OpenDeepBench: 16-dimension, 12-suite benchmark
- Correlation engine with time-series graph DB
- Multi-silicon normalized schema
- Automated root-cause diagnosis
- OpenDeepBench Composite Score

Future Work:

- Full experimental validation (11 providers)
- Agentic AI multi-step trace correlation
- CXL 3.1 memory disaggregation
- RL-based infrastructure optimization
- Industry standard composite metric

The era of single-vendor, single-metric benchmarking must end.